TANZANIA HEADS OF ISLAMIC SCHOOLS COUNCIL

FORM FOUR INTER ISLAMIC MOCK EXAMINATION

CHEMISTRY 1

MARKING SCHEME 2021

032/1

waaelimu.com

1.

I	II	III	IV	V	VI	VII	VIII	IX	X
B	D	D	D	E	C	B	A	E	E

10marks

2.

I	II	III	IV	V
B	C	I	A	D

05 marks

3. (a) In aqueous magnesium chloride; the Mg^{2+} and Cl^- ions ore formed and can easily move from one point to another. By doing so; they carry the electric charges BUT;
In solid form, there is no movement of the free ions to carry the electrical charges.
[02 marks]

(b) (i) Sodium is highly reactive metal substance with a low pulling of electrons compared to hydrogen which has high pulling of electrons
OR
Being lower in the electrochemical series, hydrogen is preferred to be selected for discharge rather than sodium. **[02marks]**

(ii) $2Cl^-(aq) \rightarrow Cl_2(g) + 2e^-$ **[01mark]**

(iii) The solution becomes alkaline due to the recombination of OH^- and $Na^+(aq)$ in the solution following the removal of Cl^- and H^+ ions. **[02marks]**

4. (a)

	Number of protons	**Number of electrons**	**Number of neutrons**
$^{39}_{19}K$	19	19	20
$^{40}_{19}K$	19	19	21

[**06marks]**

(b) Gaseous/ionized potassium
$K^+ = 2:8:8$ **[01 mark]**
K = 2:8:8:1

5. (a) Given the mass of solid potassium 0.195g
Dissolved in $500cm^3$
Volume acid $100cm^3 = 0.1dm^3$
Molarity g acid $0.1mol/dm^{-3}$.

5. (a) (i) Number of moles of hydrogen ions in 100mls of HCl.

Molarity = $\frac{number\ of\ moles}{Volume}$ **[½ mark]**

Ma = $\frac{na}{va}$

na = maxva

na = 01 x 0.1 **[01mark]**

na = 0.01 moles **[½ mark]**

(ii) Number of moles of hydroxide ions in water

$2K_{(s)} + 2H_2O(l) \rightarrow 2K^+(aq) + 20H^- + H_2(g)$

Stoichiometrically

K(s)	$OH^-(aq)$
1	1

But number of moles of potassium = mass/molar mas **½ mark**

= 0.195g/39 = **01mark**

Moles of OH^- ion before addition of acid = $5x10^{-3}$ ions. **½ mark**

(ii) Is the solution acidic or basic

$HCl + KOH \rightarrow KCl + H_2O$

1: 1

? 0.005 moles **½ mark**

Moles of HCl used = 1 x 0.005 moles

But moles of hydrogen ions added = 0.01moles

Moles of acid remaining = 0.01 moles – 0.005 moles **½ mark**

= 0.005 moles of acid

The solution will be acidic.

Because the number of hydrogen ions present in the solution will be more than of hydroxide ions. **01mark**

(b) Two things contained by a mole of a substance of any gas at s.t.p

(i) Equal volumes **½ mark**

(ii) Equal number of molecules **½ mark**

6. (a) (i) Arusha **01mark**

Because it required large volume of soap solution ($10.0cm^3$) **[01mark]**

(ii) SO_4^{2-} and HCO_3^- ions **(02 Marks)**

(b) (i) Because they contain dissolved minerals of calcium which ore essential for the development of bones and teeth. **01mark**

(ii) High amount of soap is used

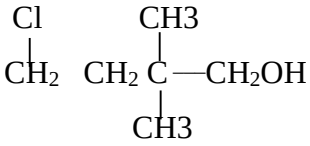
Form scum on bodies and destroy the quality of clothes

Form scales on boilers and kettles **02marks**

Blockage of sewage system due to $CaCO_3$ deposits.

Any two points

7. (a) (i) Structure

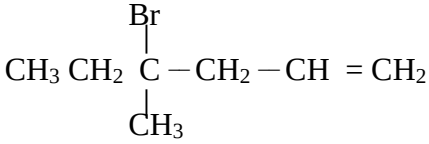


01 ½ marks

Correct 1 – chloro- 2, 2- -dimethylbutan – 1-0l. 01 ½ **marks**

Naming

(ii) Structure:

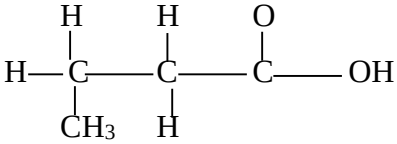


Correct : 4 – Bromo-4-methylhex-1-ene **[01 mark]**

name

(iii) Structure:

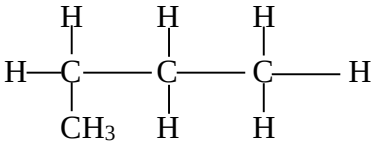
Structure:



01 mark

Correct name: Butanoic acid **01 mark**

(iv) Structure:



01 mark

Correct name: Butane **½ mark**

(b) Carbon is a chief element in organic compounds because of its ability to form a chain of its own and with other elements A property unique to itself (catenation) among all the members of group four elements. **01mark**

Partial

8. (a) (i) A ⇌ B: Reversible reaction with rapid forward reaction but very slow backward reaction. **01mark**

(ii) A ⇌ B: Reversible but dynamic equilibrium reactions where the forward and backward reaction occur at the same rate.

Irreversible **01mark**

(iii) A → B: One way non-reversible reaction where it proceeds only in one direction. **01 mark**

Burning of charcoal

(b) (i) Exothermic **½ mark**

Reason → It burns in air to produce heat (liberation of heat). **½ mark**

(ii) Photosynthesis

Endothermic **½ mark**

Reasons →It absorbs light energy for good production. **½ mark**

(iii) Evaporation of water:
Exothermic **½ mark**
Reason → Water vapour/steam is usually formed with high amount of heat released to the atmosphere. **½ mark**

(iv) Feeling cold
Exothermic **½ mark**
Reason → Loosing extra temperature gained (absorbed) when taking shower with hot water. **½ mark**

9. (a) $MnO_2(s) + 4HCl_{(l)} \rightarrow MnCl_2(aq) + 2H_2O(l) + Cl_2(q)$. **[01mark]**

(b) (i) Water→Absorption of Hydrochloric acid vapour formed alongside the chlorine gas. **[01mark]**

(ii) Concentrated sulphuric acid → Drying the gas (removal of water molecules) before it is collected by downward delivery. **01 mark**

(c) Because the gas is more denser than air…….. **01 mark**

(d) Uses of sulphuric acid:
(i) Manufacture of paints and pigments
(ii) Catalyst in petroleum refining
(iii) Manufacture of soap and detergents **03marks**
(iv) In lead-acid car batteries (accumulators)
(v) In manufacture of fertilizers
(vi) Purification of gold
Any 03 points @ 01 mark

10. (a) Raw materials in iron production.
(i) Iron ore (Haematile) →Chief source of iron. **01 ½ marks**
(ii) Calcium carbonate (marble chips)→Removal of impurities **01 ½ mark**
(iii) Coke (carbon) → Reduction of iron form us oxide. **01 ½ mark**
(iv) Hot air → oxidation of coke to its oxide energy
Oxidation of ore to ore-oxide **1 ½ marks**

(b) The walls are made up of calcium oxide has got high heat resistance. The more is heated the more it becomes inactive. (High melting point **01mark**

11. (a) Data given
Mass of a gas = 0.8g
Volume g a gas = $560cm^3$ at s.t.p = $0.56dm^3$ **½ mark**
Required: Molar mass of a gas
Solution
At s.t.p each one mole g a gas occupies a total volume g $22.4dm^3$=$22400cm^3$.
½ mark
$\therefore$ $0.8g \rightarrow 0.56dm^3$ **½ mark**
$? \rightarrow 22.4dm^3$

$\frac{0.8g \times 22.4dm^3}{0.56}$ **01 mark**

The molar mass of a gas is 32g/mol **½ mark**

The gas is O_2. **0 ½ mark**

(b) (i) Y → Is a carbon dioxide gas (CO_2) **½ mark**

A gas is tested by lime water ($Ca(OH)_2$) which changes milky it is bubbled into it. **½ mark**

(ii) Z → Is water (H_2O) **½ mark**

→ It can be tested in three ways

- Changes cobalt chloride paper from blue to pink
- Turns an hydrous copper(II) sulphate to blue. **01mark**
- Form an alkaline solution with Na and K. Which turns litmus paper blue.

(iii) $CaCO_3(s) + 2HNO_3(aq) \rightarrow Ca(NO_3)_2 + CO_2(g) + H_2O(l)$ **01mark**

12. (a) (i) Because in dilute HCl copper cannot displace hydrogen from mineral acids as it is less reactivity compared to hydrogen. But in conc. Nitric acid copper is oxidize to copper nitrate because $HNO_3(l)$ is a strong oxidizing agent. **01mark**

(ii) Because conc. H_2SO_4 is hydroscoping compound that withdraws all elements of water from the compound. **01mark**

(iii) Because egg shells contains calcium carbonate which decomposes to CO_2 and corresponding salts on addition of HCl. **01mark**

(iv) Because silver nitrate is highly sensitive to just litre amount/of trace of chlorine ions to form silter chloride solids. **01mark**

(b) (i) Sodium hydroxide plays role of identifying a presence of any insoluble metal hydroxides which will precipitate the (OH^-) ion from sodium. If no precipitate is formed, means the salt can probably contains sodium or ammonium salt. **01 ½ marks**

(ii) Barium chloride → Is for confirming the presence of sulphates in any given salts. For it forms a precipitate of barium sulphates. **01 ½ marks**

SECTION C

13. Introduction

Meaning of global warming → Un usual increase in the average temperature of the Earth's atmosphere as a results of greenhouse effects. **01mark**

Greenhouse effects: Is the trapping and retainment of heat energy which would otherwise escape and instead is reflected back to the Earth's surface and results to over increase of the Earth's surface's temperature. **02marks**

Effects:

(i) Melting of ice in polar regions. **(Point 01 mark. Explanation 01 mark)**

(ii) Floods and rise in sea level **Any 3 points @ o2**

(iii) Climatic changes such as heavy rainfall, expansion of deserts, storms, typhoons, hurricanes. **06**

(iv) Loss of biodiversity due to less resistance to rising temperature.

(v) Increase in tropical diseases such as malaria in malaria free zones.

Prevention

(i) Encourage the use of environmentally friendly sources of energy (nuclear energy for example) **06**

(ii) Proper ways of waste management such as recycling instead of burning

(iii) Encourage reforestation and afforestation and discourage deforestation

(iv) International treaties to minimize the emission of greenhouse gases

(v) Public transport to be encouraged **Any 3 points @ 2 marks**

14. Introduction

(i) Meaning and categorization of plants nutrients
- Macro nutrients → required in large proportion
- Micro nutrients → required in large proportion

03 Marks

(ii) Primary macro nutrients

N
- Formation of protein, enzymes hormones
- Control availability of K and P.
- Plant growth

any one point 02

P
- Strengthening plant stems
- Formation g root nodules in legumes
- Formation of flowers, roots and seeds
- Disease resistance
- Synthesis of ATP, ADP, GTP (energy rich compounds)

Any one point 02

K
- Plants photosynthesis
- Translocation of carbohydrates
- Cell fungo pressure and permeability
- Disease resistance
- Neutralization of organic acids

Any one point 02

(iii) Management of loss of plants nutrients from the soil
- Prevention of soil and erosion
- Crop rotation
- Mixed cropping
- Good harvesting method
- Mulching
- Addition of manure

Any three points 06